

Thiết kế giao diện

CONTROLS

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1. User

1. Introduction
2. Human information process model
3. *Cognitive process*

2. Usability

1. Concept
2. Usability components
3. Usability design guidelines

3. Class project [1]

1. Input & Output design

2. Controls

1. Introduction
2. Some typical control types
3. User Controls

Input design & Output design

✦ Input design:

- Input entity data (data list): e.g. book, customer, product...
- Input relational data: e.g. book borrow, sale order...
- Input search

✦ Output design:

- Output to screen
- Output to printer
- Output to file
- Output search result

Input design

- ✦ Input decides the correctness of systems
- ✦ Two (2) types of input:
 - ***Real-time input***: use form & controls
 - Batch input: ~ import mass of structured data
- ✦ The best input design is the simplest & most natural one.
- ✦ Error prevention & Data validation

Input design

Input design guidelines:

- ✦ Match the forms to real source documents
- ✦ Input necessary data only
- ✦ Have default value
- ✦ Display a list of values for selectable fields
- ✦ Have correct tab order
- ✦ Provide descriptive labels for every fields
- ✦ Provide examples for special requirement (e.g. special format)
- ✦ Always allow to exit the input screen without saving

Output design

- ✦ Different output: screen, printed paper or both.

- ✦ Output to screen:
 - Especially grid (table) control
 - Data binding

- ✦ Output to printed paper: report
 - Report tools; e.g. Crystal Report, Jasper Report...

Output design

Report design:

- ✦ Based on existing reports
 - ✦ Should be evaluated by users as soon as possible
 - ✦ Notice report header/footer & page header/footer
 - ✦ Spacing is important
 - ✦ Group information
 - ✦ Uniform & consistent
- Many powerful report tools

1. Input & Output design

2. Controls

1. Introduction

2. Some typical control types

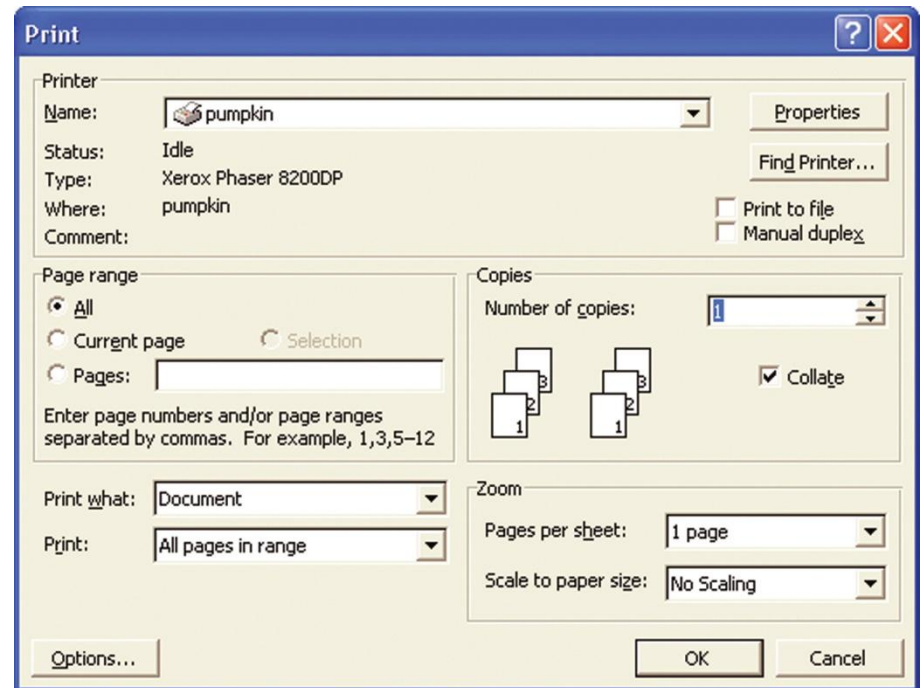
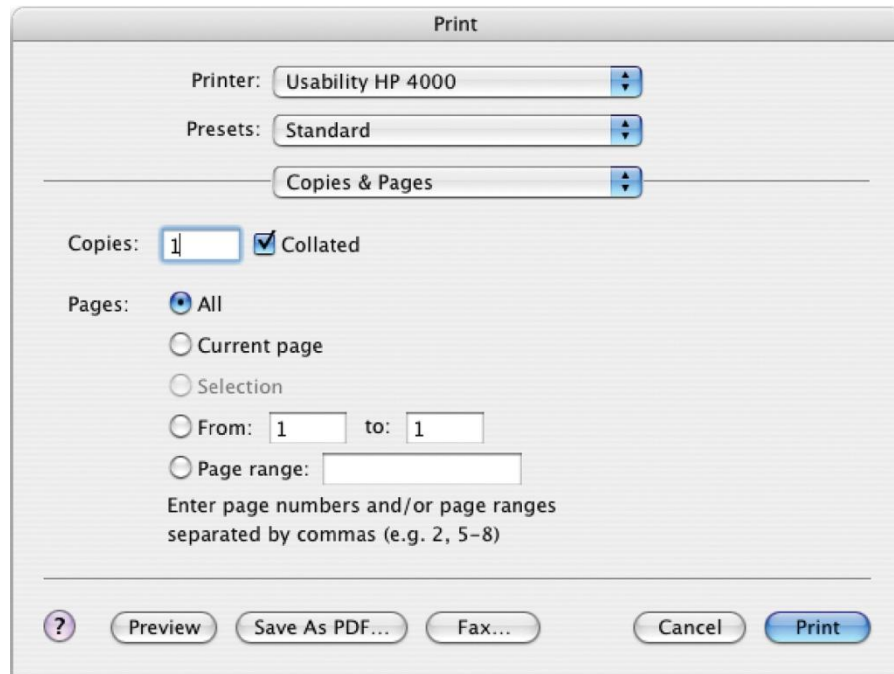
3. User Controls

Controls

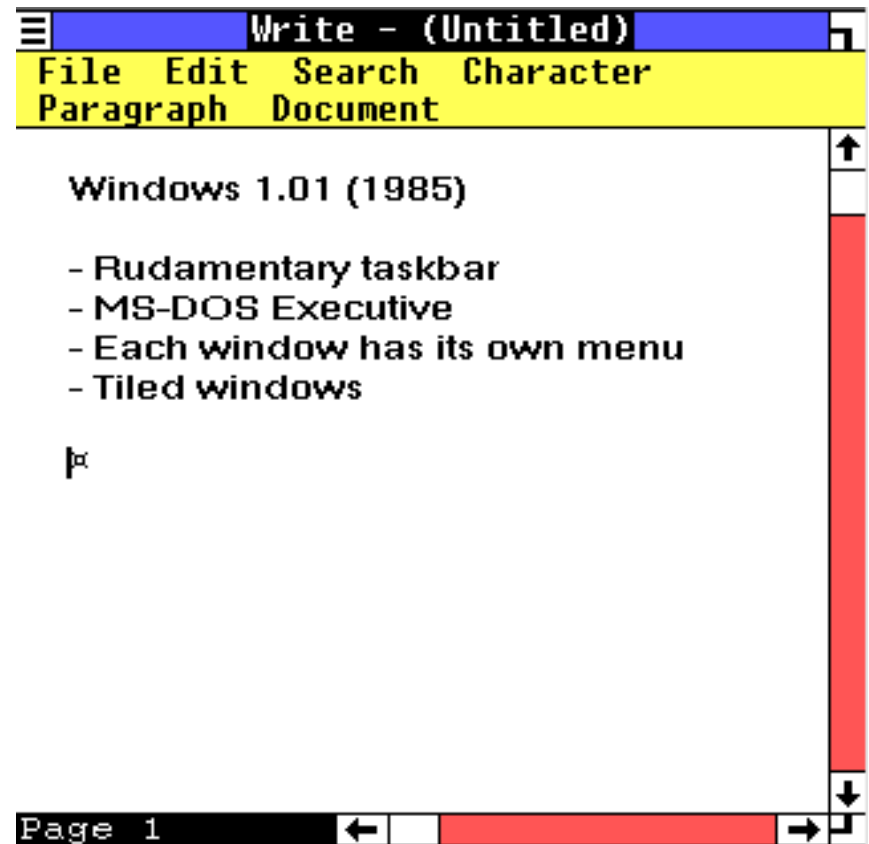
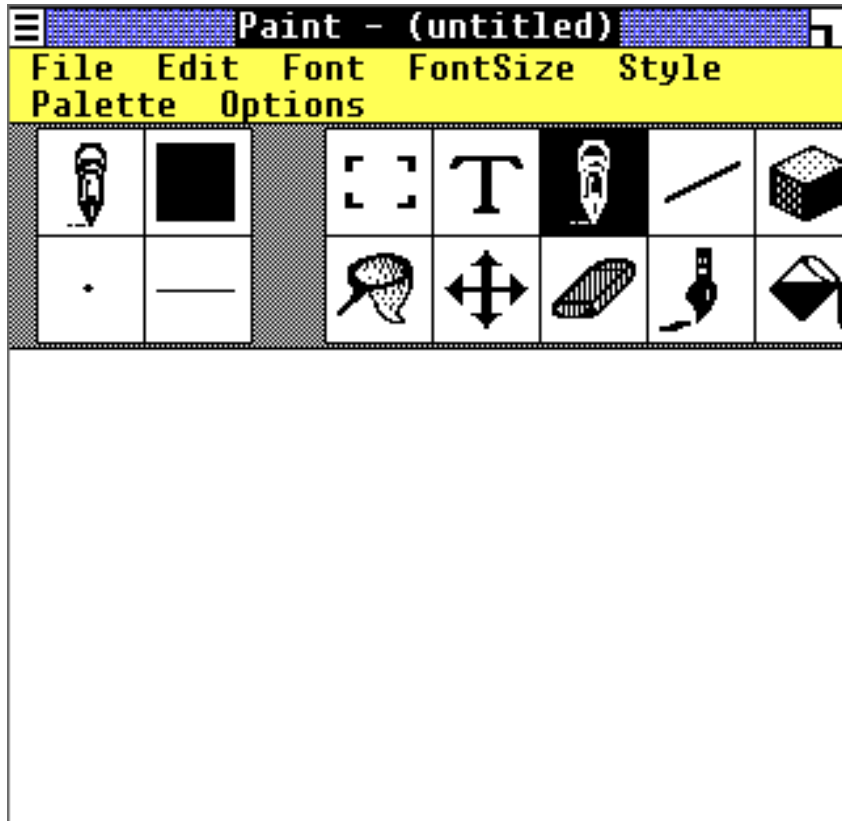
- ✦ Also called widgets
- ✦ Reusable user interface components
- ✦ Handle both input & output
- ✦ Event handling: callback function (delegate, listener)
- ✦ Set of controls have different style. E.g. XP, Vista, MAC style...

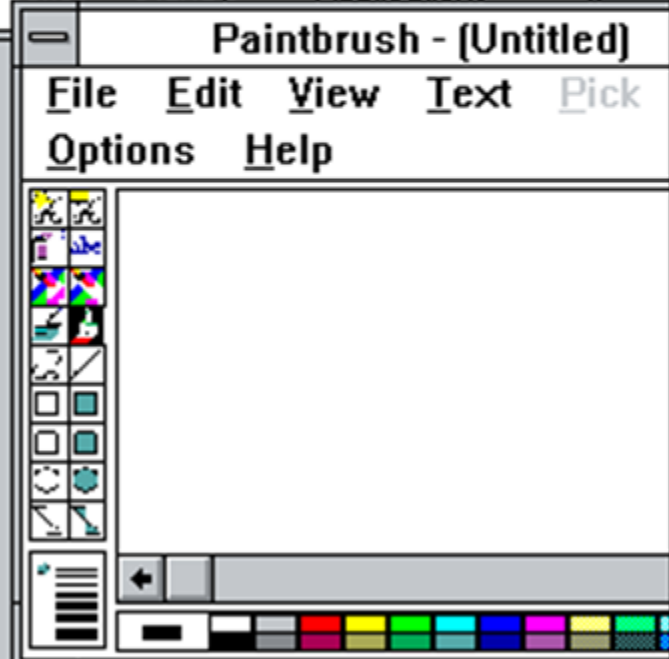
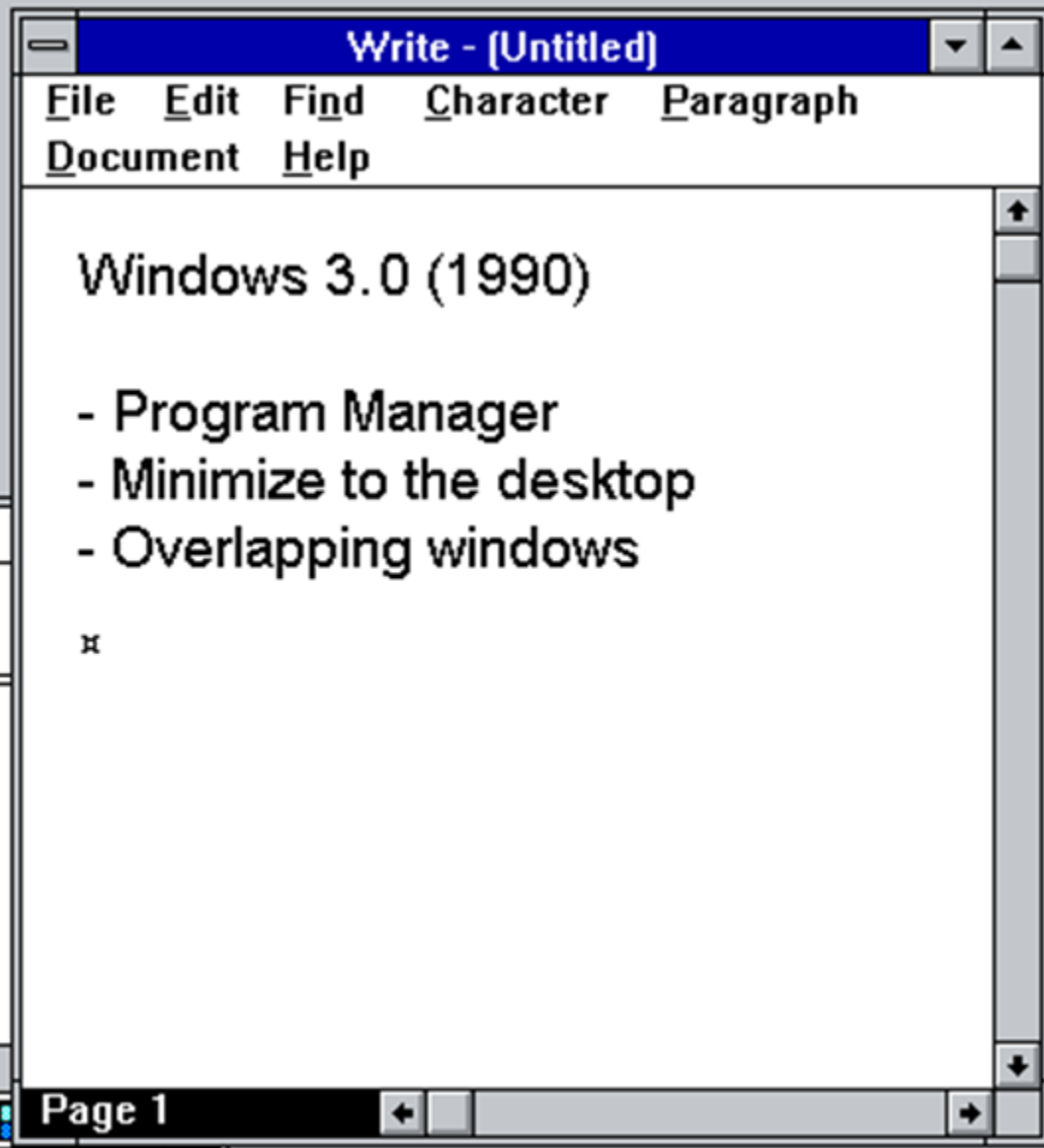
Controls

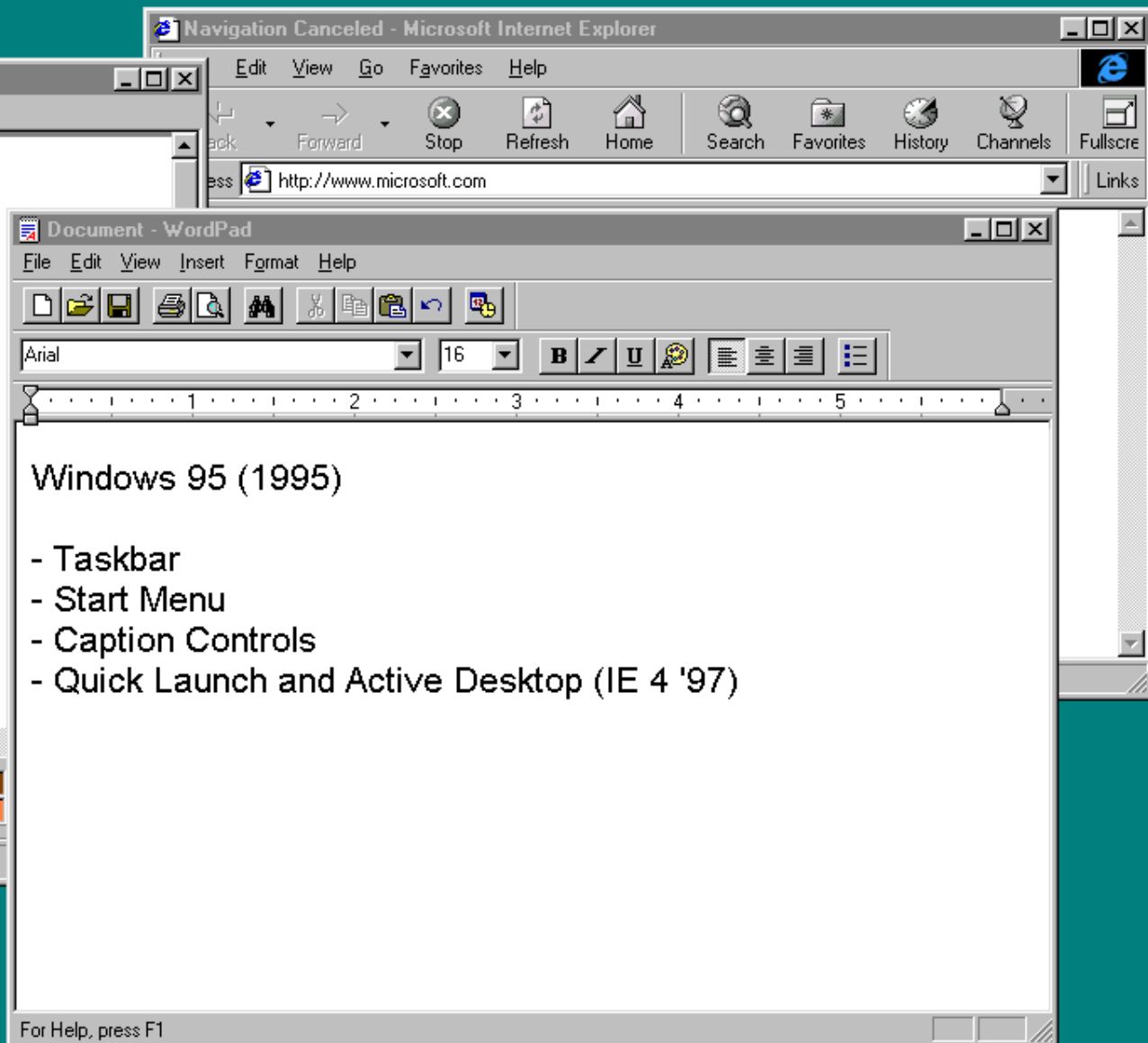
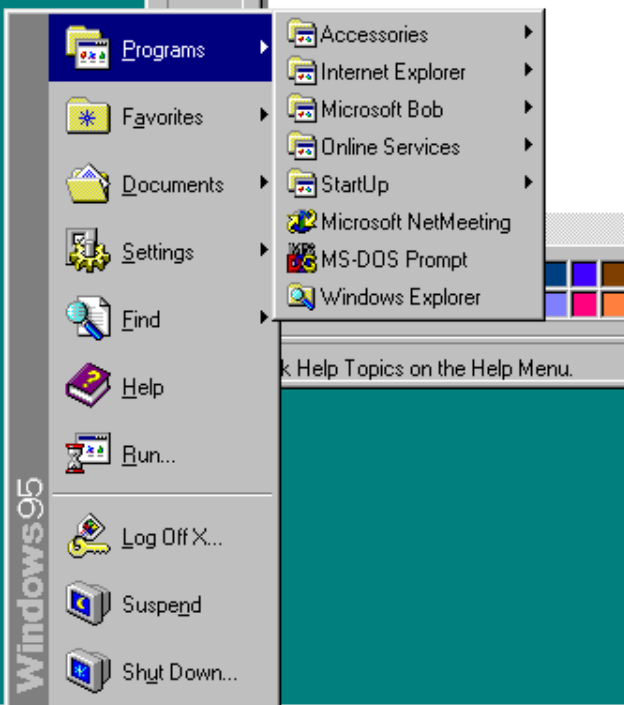
Controls on MAC OS and Wins XP OS

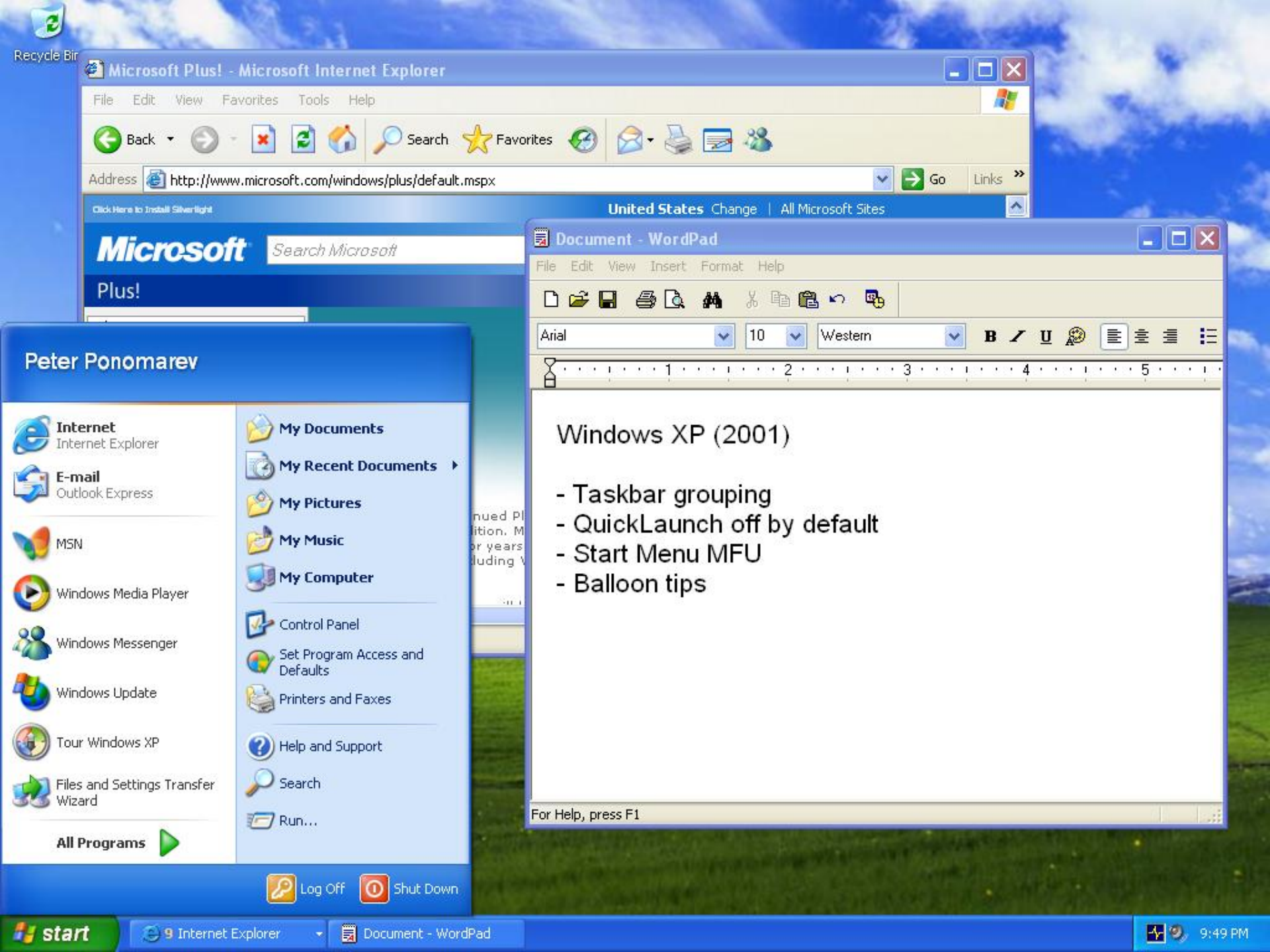


Controls – Windows GUI Evolution









Recycle Bin

Microsoft Plus! - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites Refresh Print Mail Stop People

Address <http://www.microsoft.com/windows/plus/default.mspx> Go Links

Click Here to Install Silverlight

Microsoft

Search Microsoft

Plus!

United States Change | All Microsoft Sites

Document - WordPad

File Edit View Insert Format Help

New Open Save Print Find Copy Paste Undo Redo

Arial 10 Western Bold Italic Underline Bulleted List Numbered List Indent Decrease Indent Increase

1 2 3 4 5

Peter Ponomarev

Internet
Internet Explorer

E-mail
Outlook Express

MSN

Windows Media Player

Windows Messenger

Windows Update

Tour Windows XP

Files and Settings Transfer
Wizard

All Programs

My Documents

My Recent Documents

My Pictures

My Music

My Computer

Control Panel

Set Program Access and
Defaults

Printers and Faxes

Help and Support

Search

Run...

Windows XP (2001)

- Taskbar grouping
- QuickLaunch off by default
- Start Menu MFU
- Balloon tips

For Help, press F1

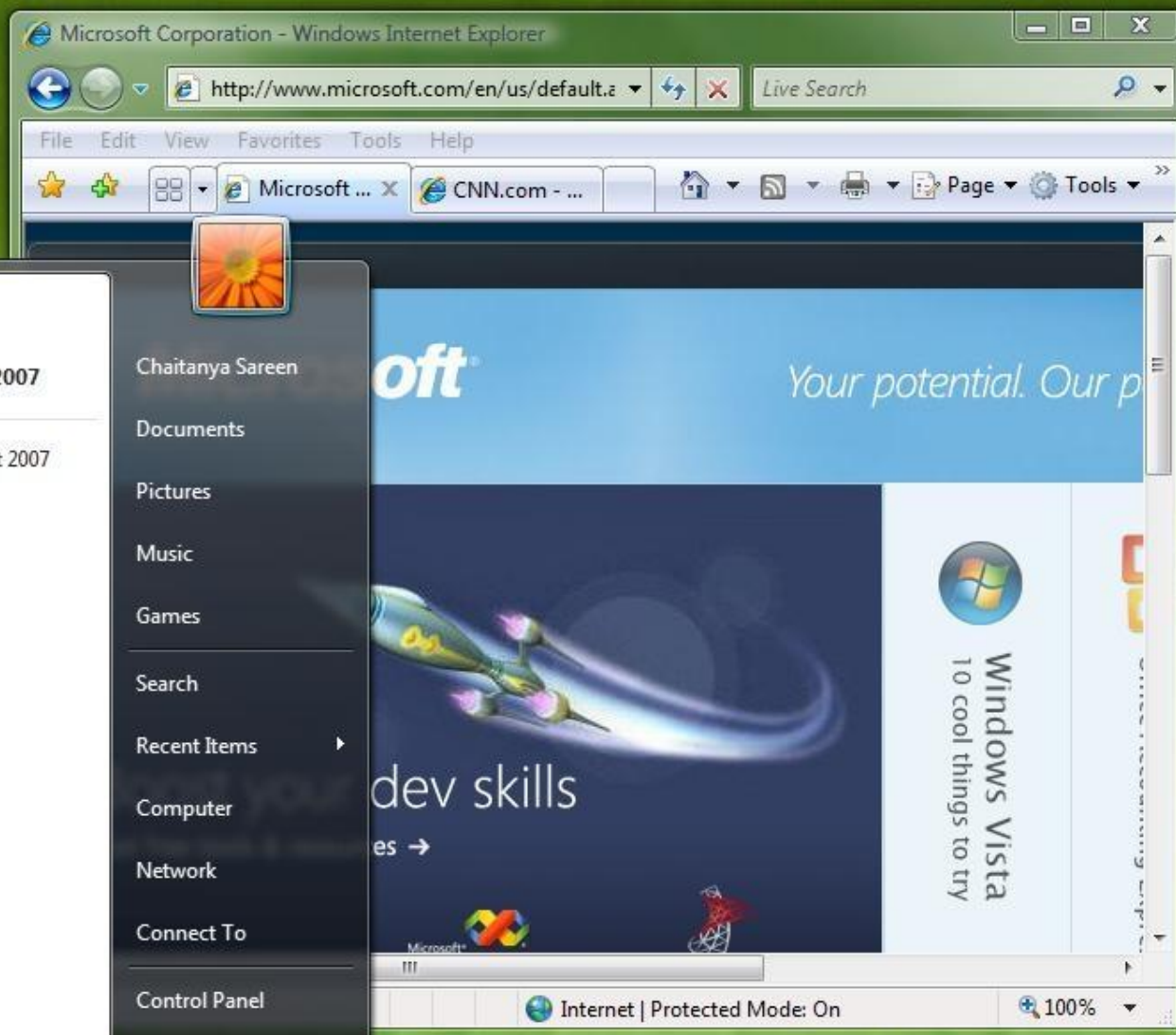
Log Off Shut Down

start

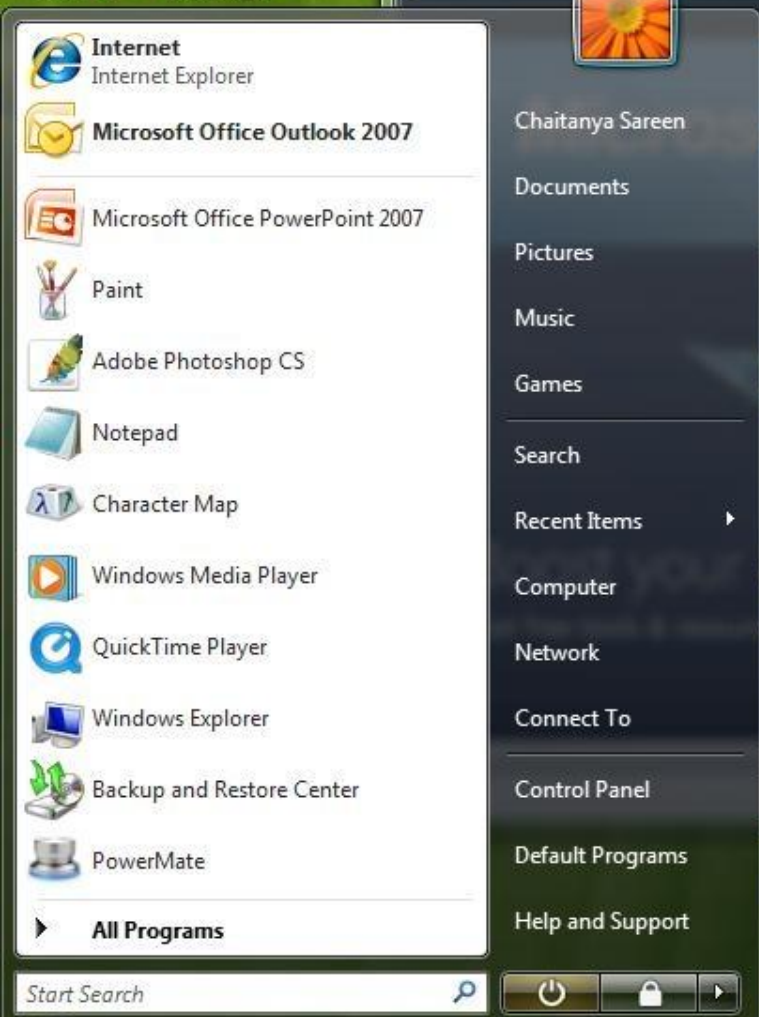
Internet Explorer

Document - WordPad

9:49 PM

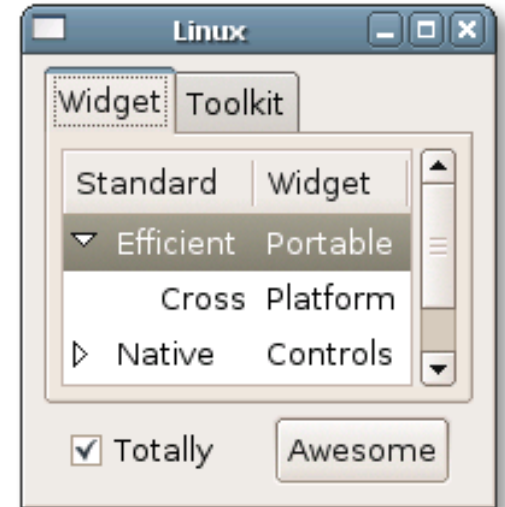
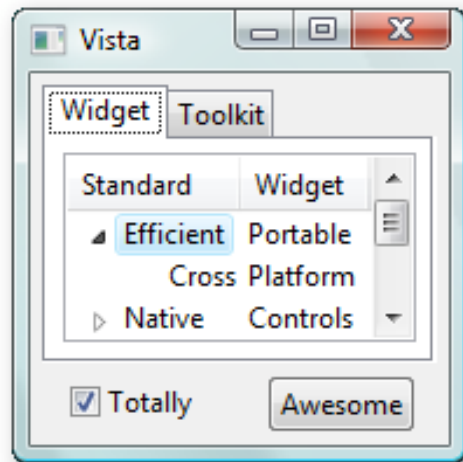


How many
Outlook icons
do you see?



Controls

Different style controls



Controls

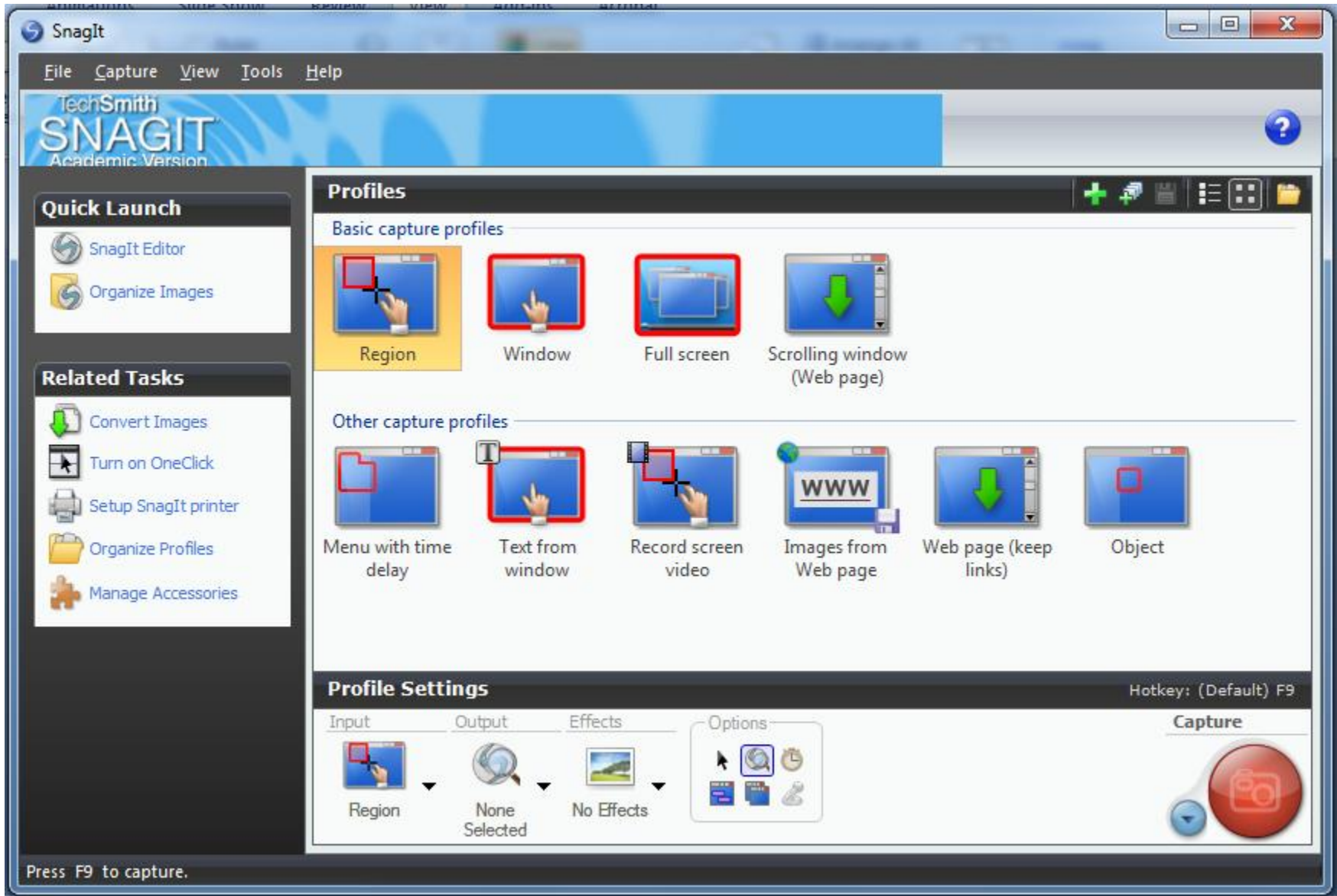
- ✦ Programming languages provide common & standard control classes:
 - C#: System.Windows.Forms, ...
 - Java: awt, swing, swt (eclipse)...
- ✦ Properties, methods, event handlers of control classes have same basis but different syntax & features.
- ✦ Inherit (derive) control classes

Controls

- ✦ SDI: Single Document Interface: one window acts as an interface. E.g. Window Explorer, Notepad
- ✦ MDI: Multiple Document Interface: at least one parent and many child windows. E.g. Ms. Word, Firefox

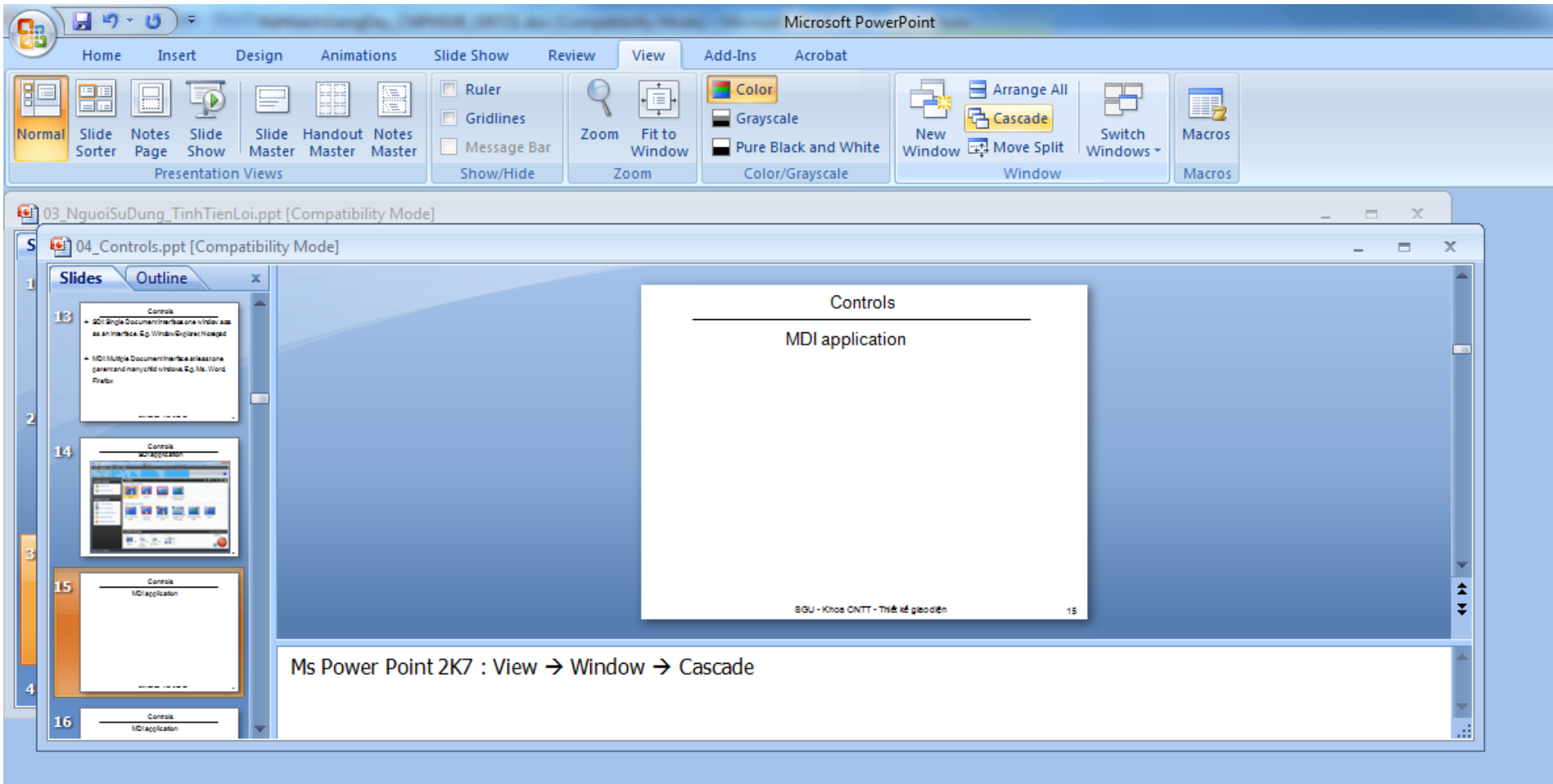
Controls

SDI application



Controls

MDI application



Controls

MDI application

VnExpress - Daily News - Mozilla Firefox

File Edit View History Bookmarks Yahoo! Tools Help

http://vnexpress.net/GL/Home/

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user control in c sharp tutorial - Go... VnExpress - Daily News

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Trang nhất Xã hội Thế giới Kinh doanh Văn hóa Thể thao Pháp luật Đời sống Khoa học Vì tính Ôtô - Xe máy Ba

RSS Liên hệ tòa soạn Đặt làm trang chủ Ngôi sao Số hóa Game thủ Việc làm Eba

Thứ hai, 26/7/2010 | 15:58

Lễ hội âm nhạc 'là ác mộng'

"Tôi thấy 25 người chết đồng loạt. Tôi thấy một người xanh lét và muốn cho anh ít nước, nhưng anh ấy chết rồi", một người tham dự Love Parade ở Đức, nơi 18 người chết vì bị giẫm đạp, kể lại.

> 18 người chết vì bị giẫm đạp ở Đức

- Cú sút penalty siêu quái của tài năng Tây Ban Nha
- 'Vua bếp' Yan Can Cook nấu phở Việt Nam
- Cận cảnh 11 căn nhà bị 'nuốt chửng' dưới sông
- Hơn 80 tỷ đồng vi phạm tài chính Đảng được phát hiện
- 2 CĐV Hải Phòng bị tạm giữ vì gây rối trên sân Hàng Đẫy
- Đại gia chứng khoán, địa

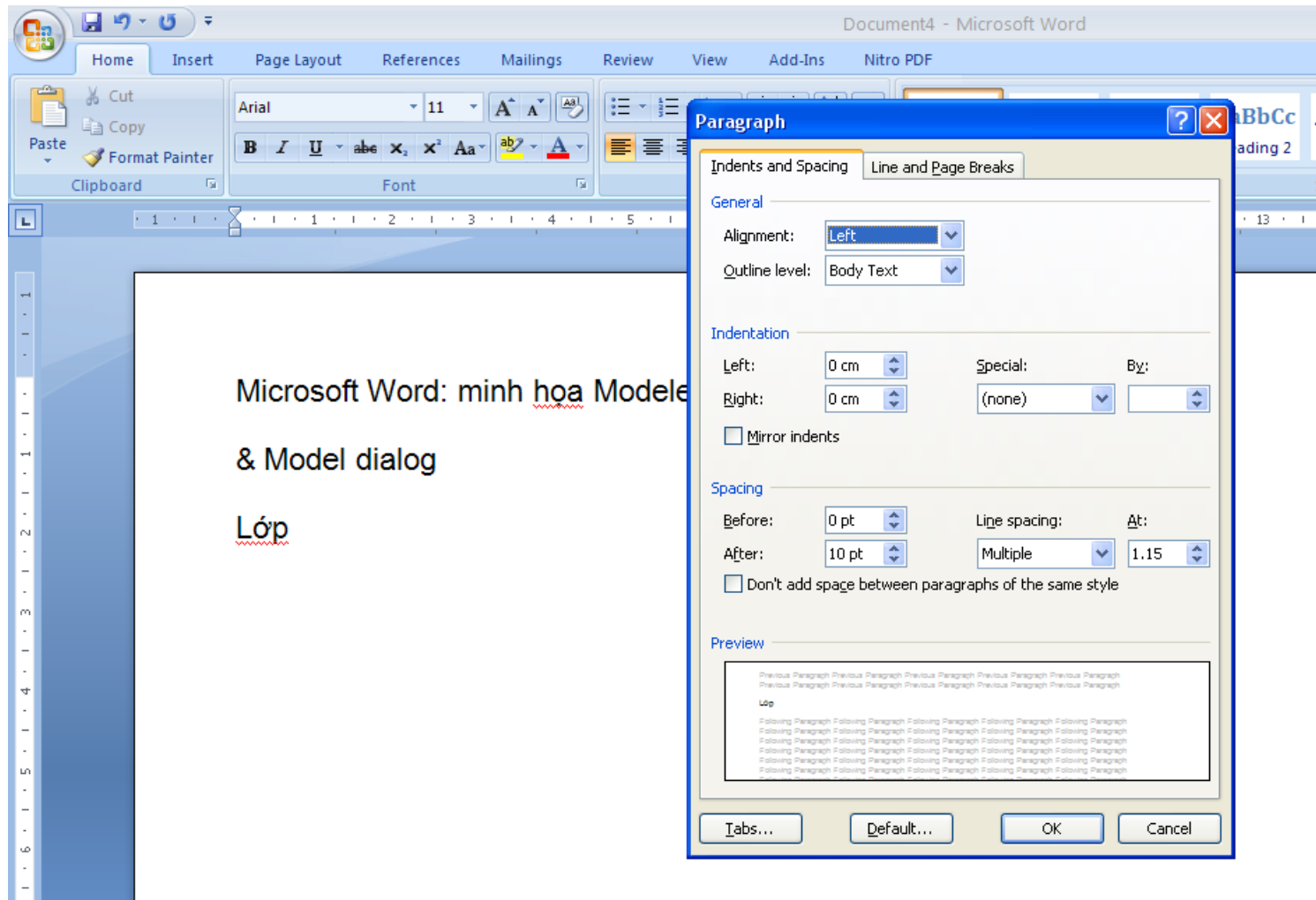
QUẢNG CÁO

Sacombank

Phiếu quà tặng

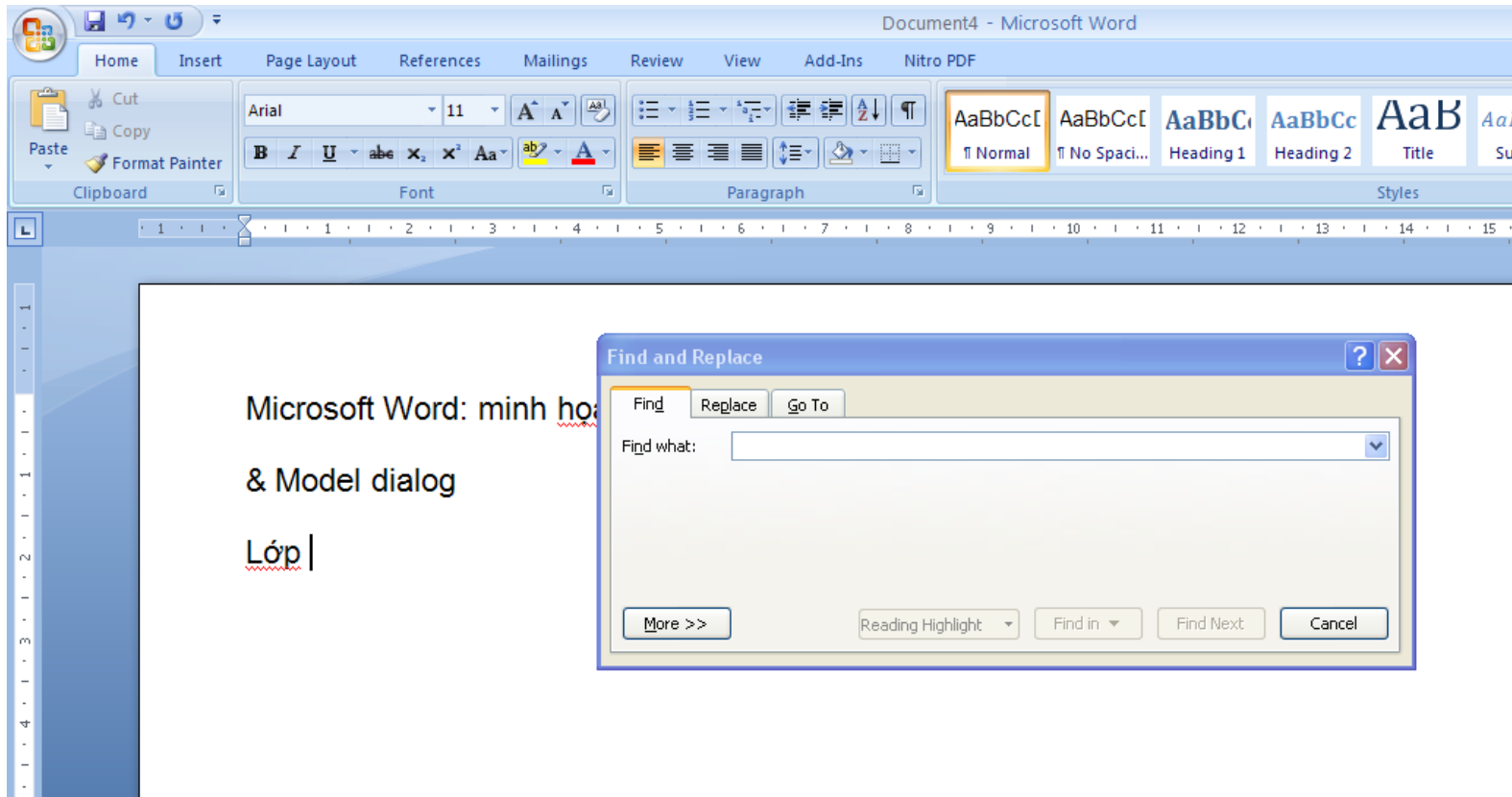
Controls – for container

Dialog (form): modal



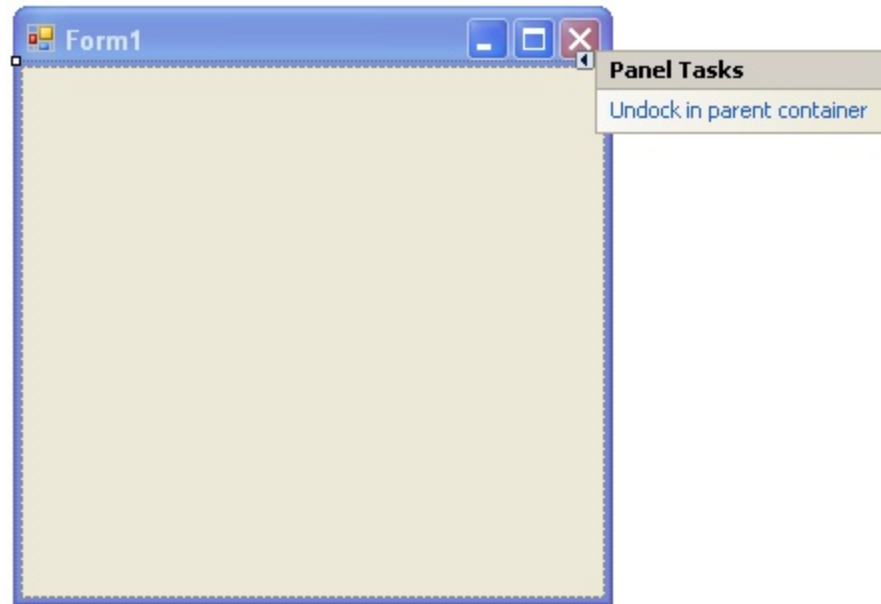
Controls – for container

Dialog (form): modeless

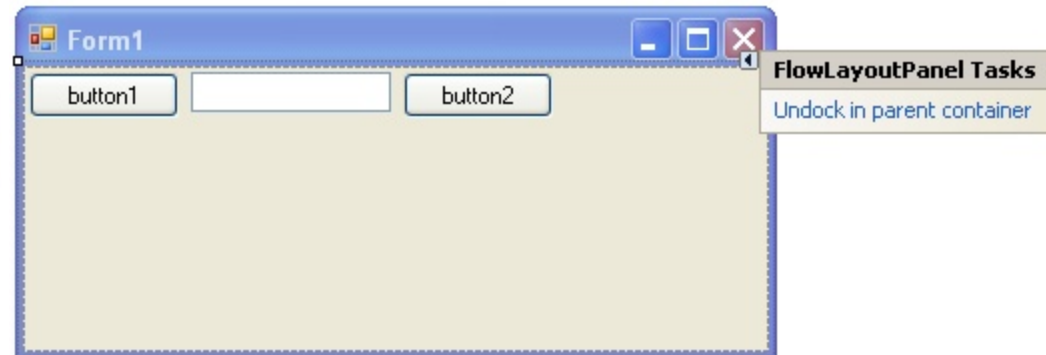
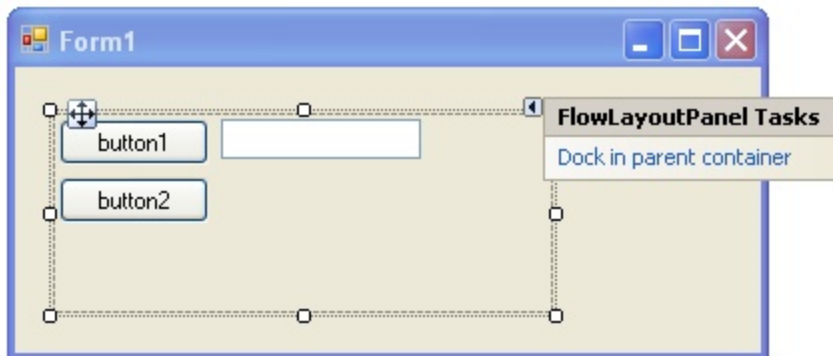


Controls – for container

Panel

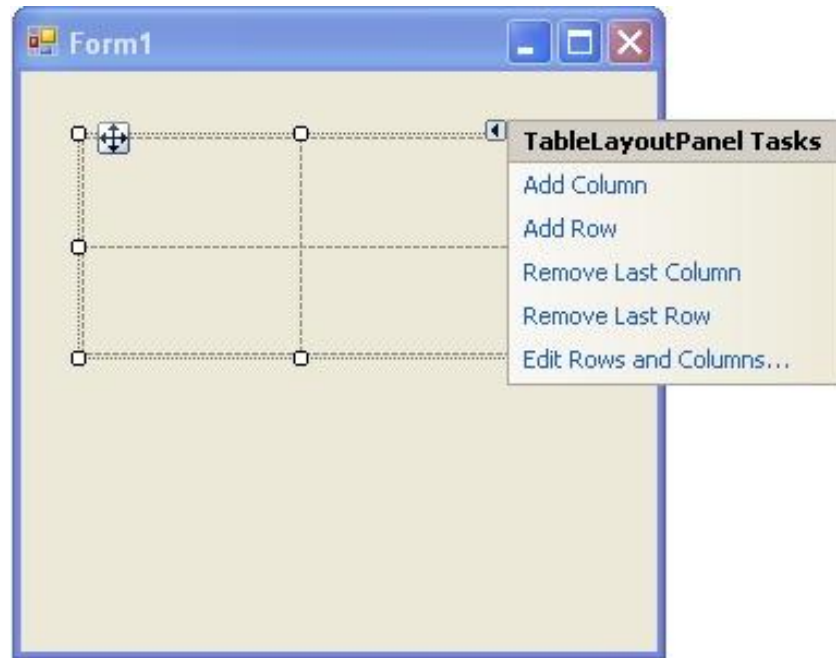


FlowLayoutPanel



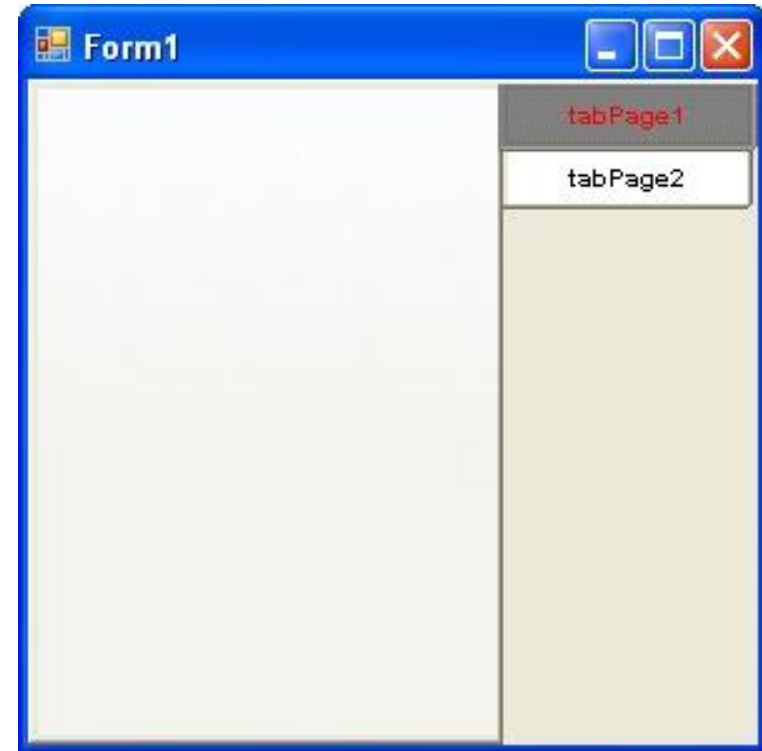
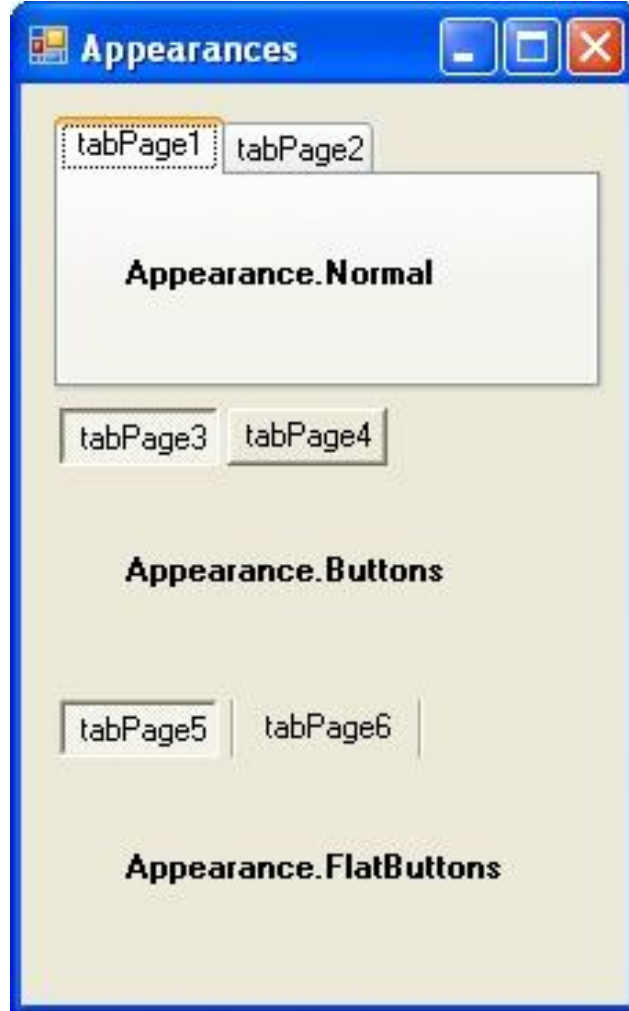
Controls – for container

TableLayoutPanel



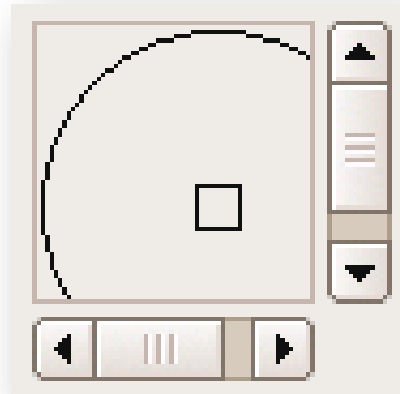
Controls – for container

TabControl



Controls – for organize content

Scrollbar

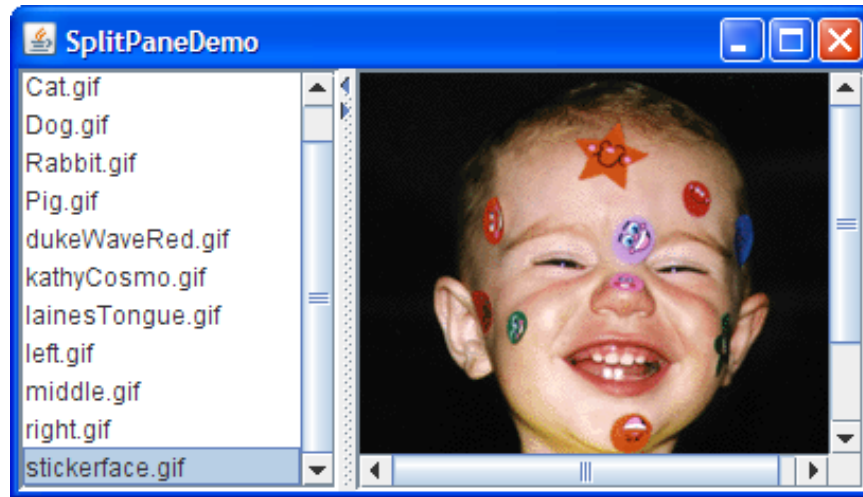


Groupbox

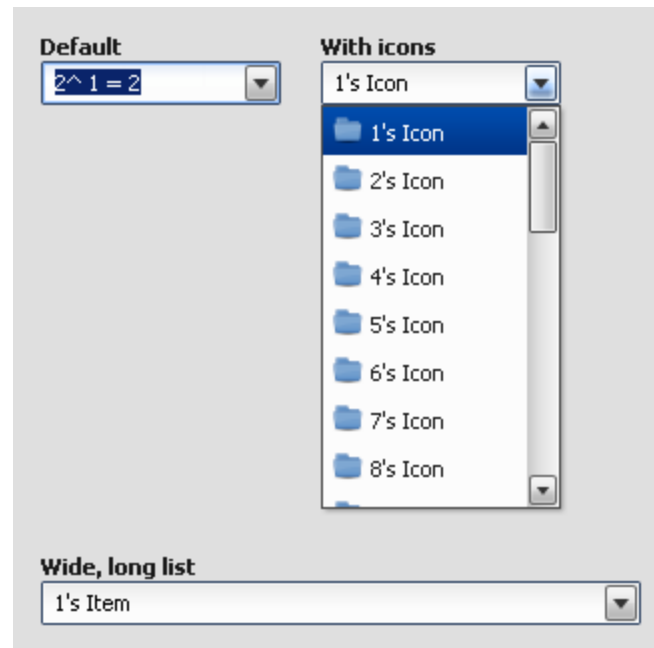


Controls – for organize content

Listbox

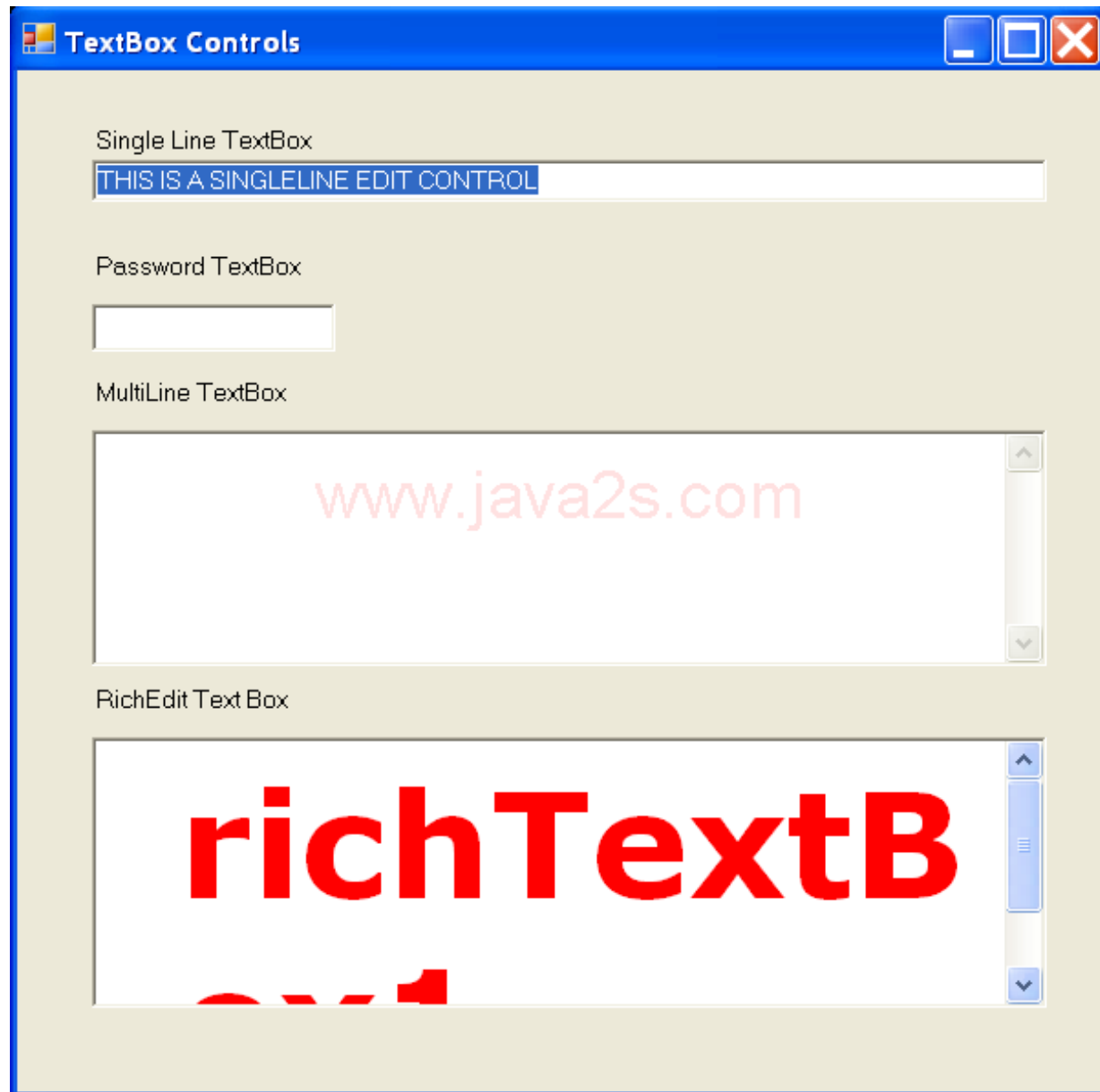


Combobox



Controls – for text

Textbox

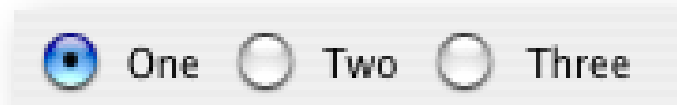


Controls – for choices

Checkbox

- ☒ Lettuce
- ☐ Tomato
- ☒ Onion
- ☐ Pickle

Radio button

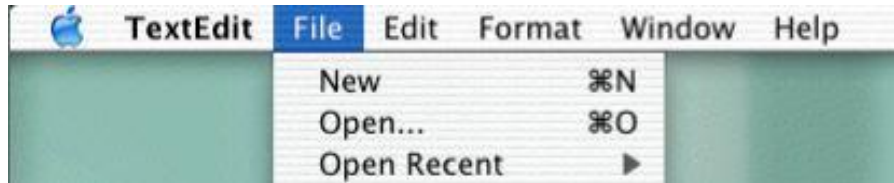


Toggle button

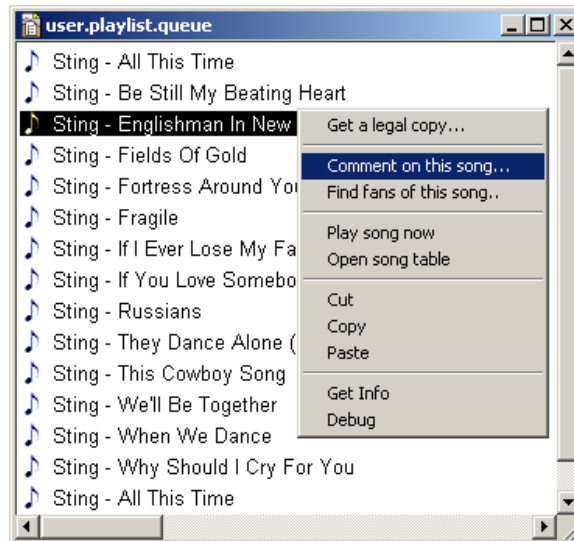


Controls – for commands

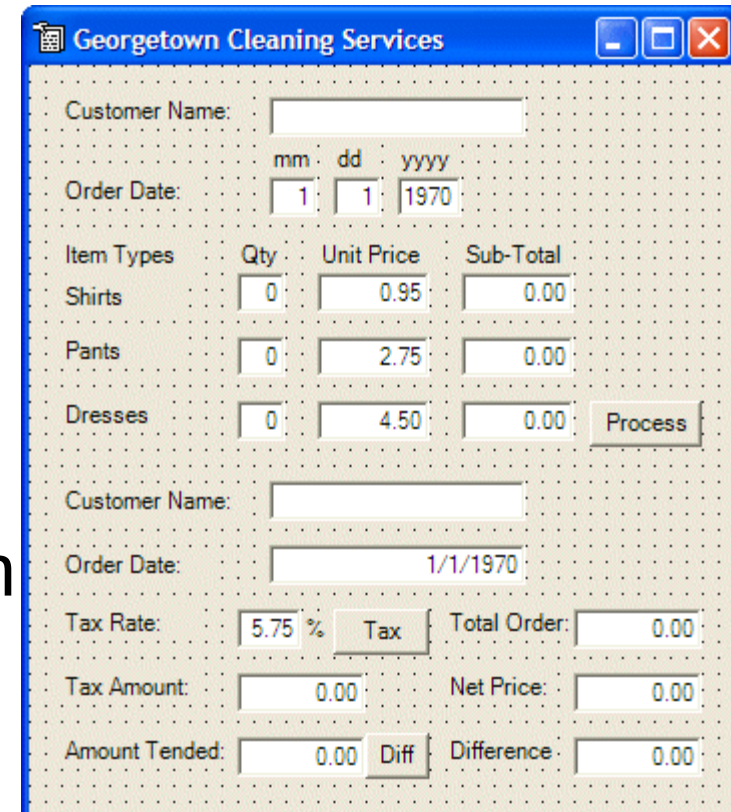
Menu



Context Menu



Button

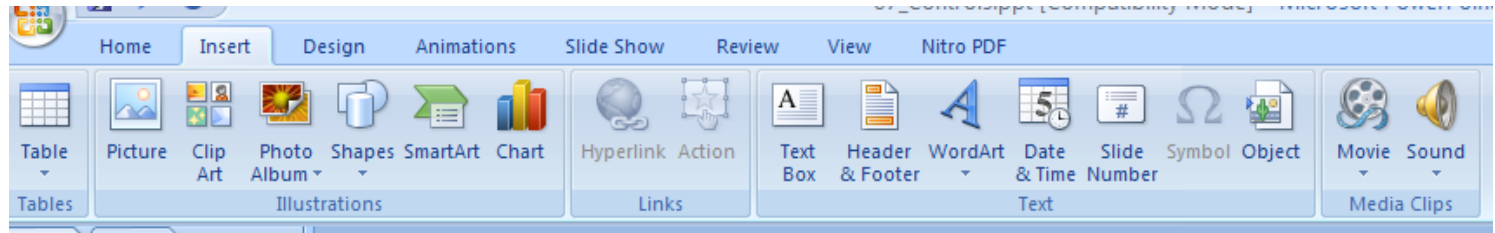
A screenshot of the 'Georgetown Cleaning Services' application window. It features a form with various input fields and buttons. The form includes sections for Customer Name, Order Date (mm/dd/yyyy), Item Types (Shirts, Pants, Dresses) with Qty, Unit Price, and Sub-Total columns, and a 'Process' button. Below this, there are fields for Customer Name, Order Date (1/1/1970), Tax Rate (5.75%), Tax, Total Order (0.00), Tax Amount (0.00), Net Price (0.00), Amount Tended (0.00), Diff, and Difference (0.00).

Controls – for commands

Toolbar



Ribbon



Hyperlink



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Controls – for others

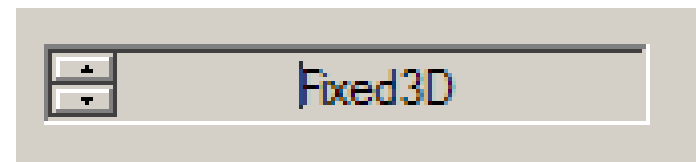
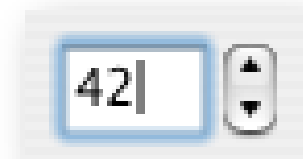
DateTimePicker



TreeView

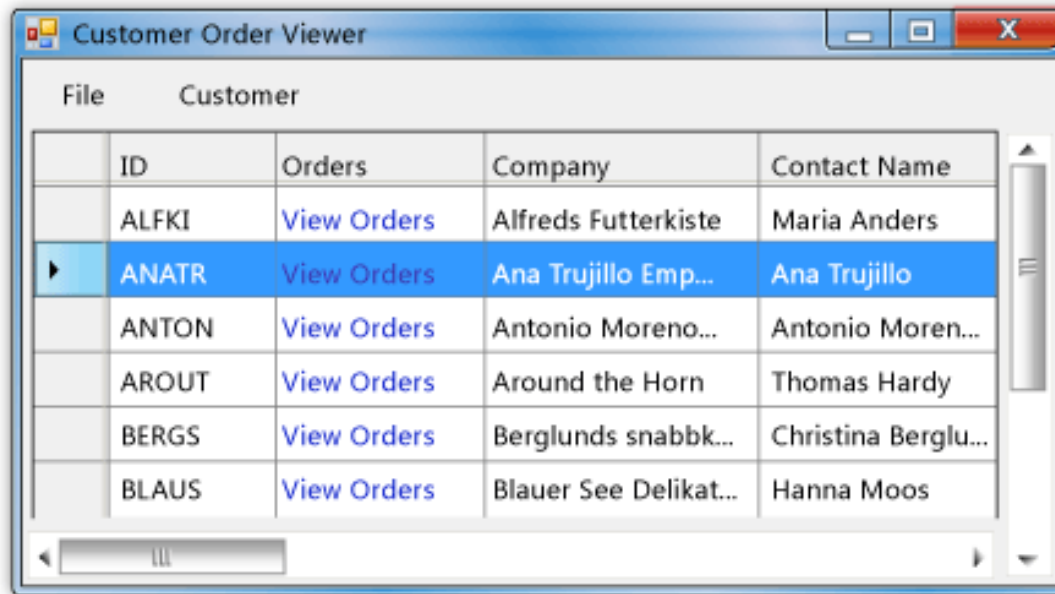


Spinner
(DomainUpDown)



Controls – Datagrid

DataGrid
(Table)



Name	Type	Size
☐ Index:0	classes	0
☒ Index:1	databases	2556
☐ Index:2	images	9157
☒ Index:3	classes	0
☐ Index:4	databases	2556

User control (i.e. custom widget)

✦ **Inherit** standard control classes to provide better display style & more features:

- Java: usually free widgets
- .NET: mostly commercial control sets.
 - DevExpress Winform (from \$799)
 - Telerik RadControls for WinForms (from \$799)
 - Infragistics NetAdvantage® for Win Client (from \$995)

Simple icons plus labels



Delete



Redo



Undo



Properties



Cut



Copy



Paste



Folder Options



Views



Back



Forward



Stop



Refresh



Home



Search



Favorites



History



Mail



Up



Move To



Copy To



Folders



Open



Save



Print

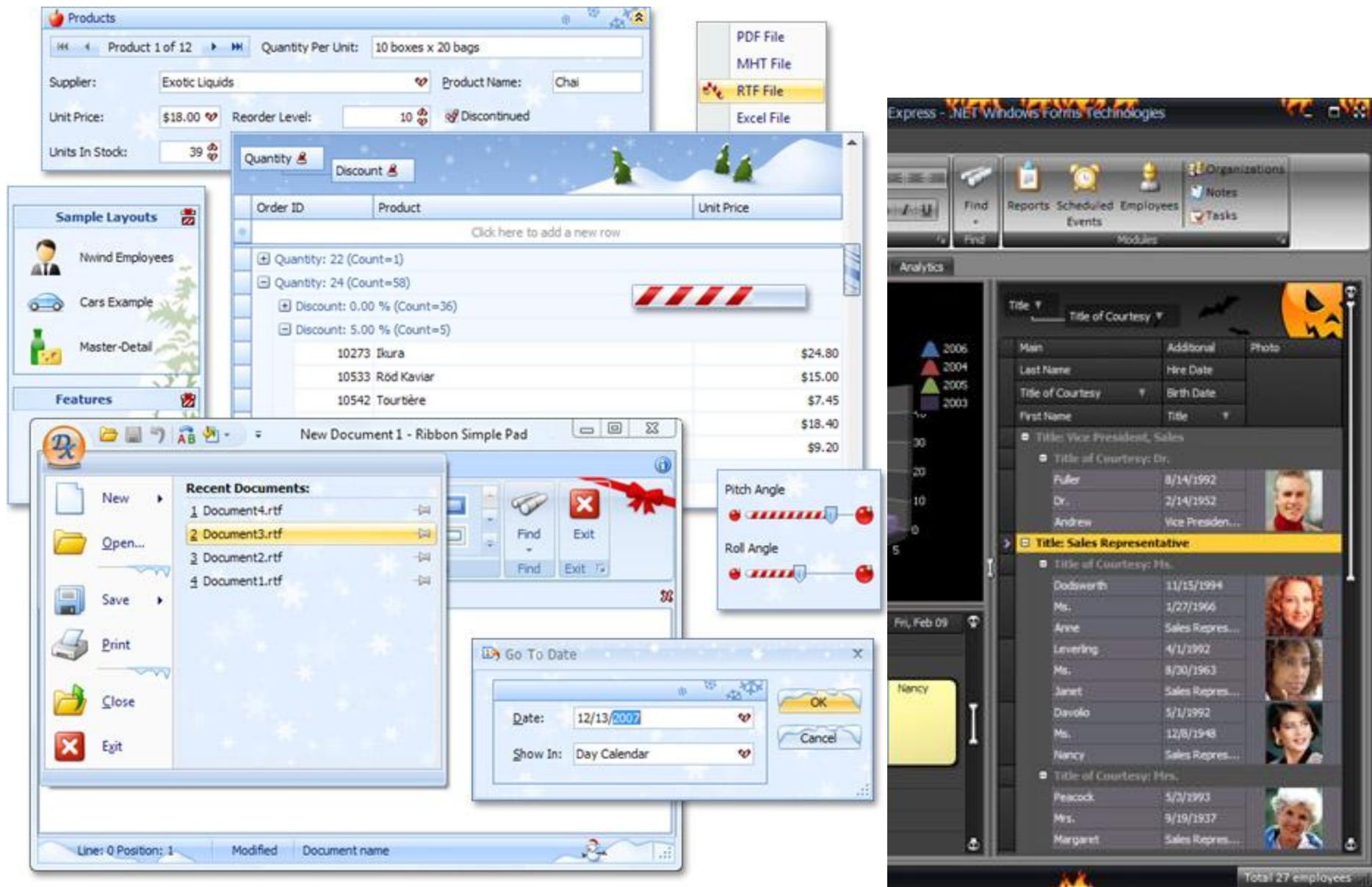


New

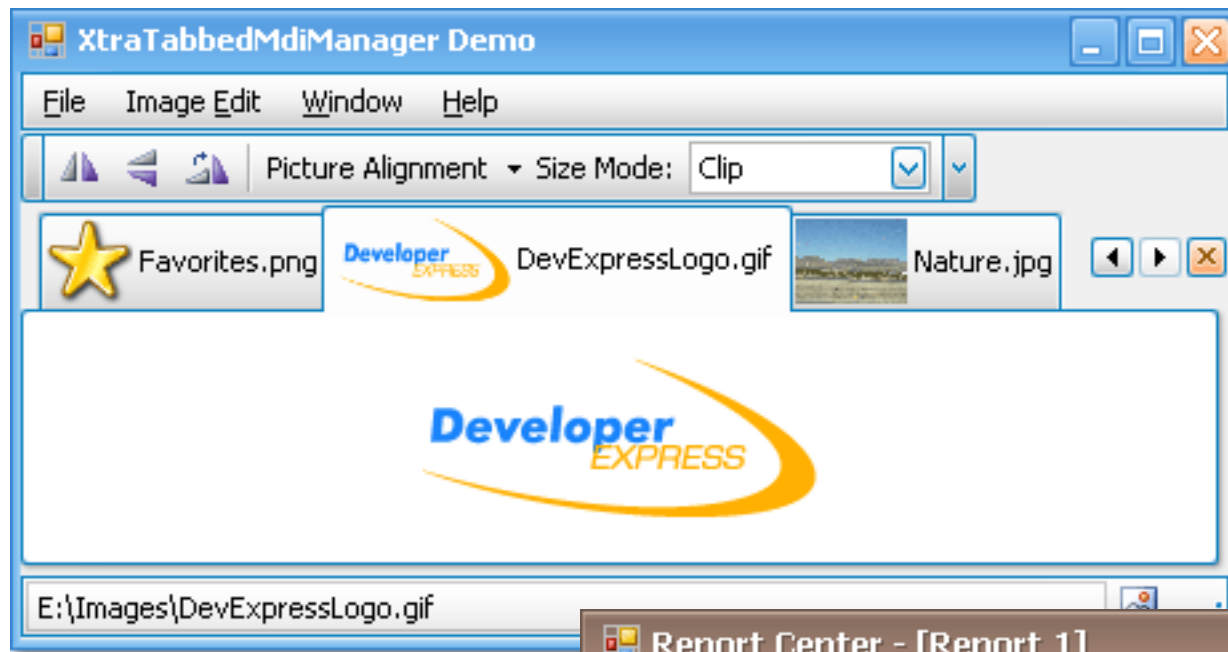


Print Preview

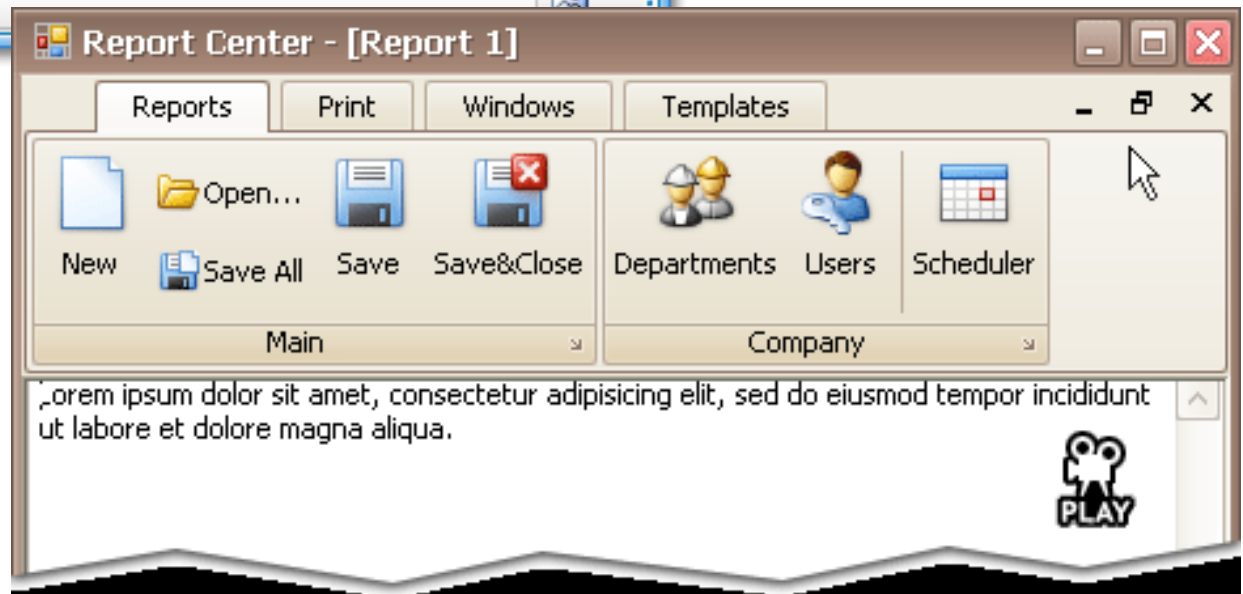
Controls – DevExpress Winform



Controls – DevExpress Winform

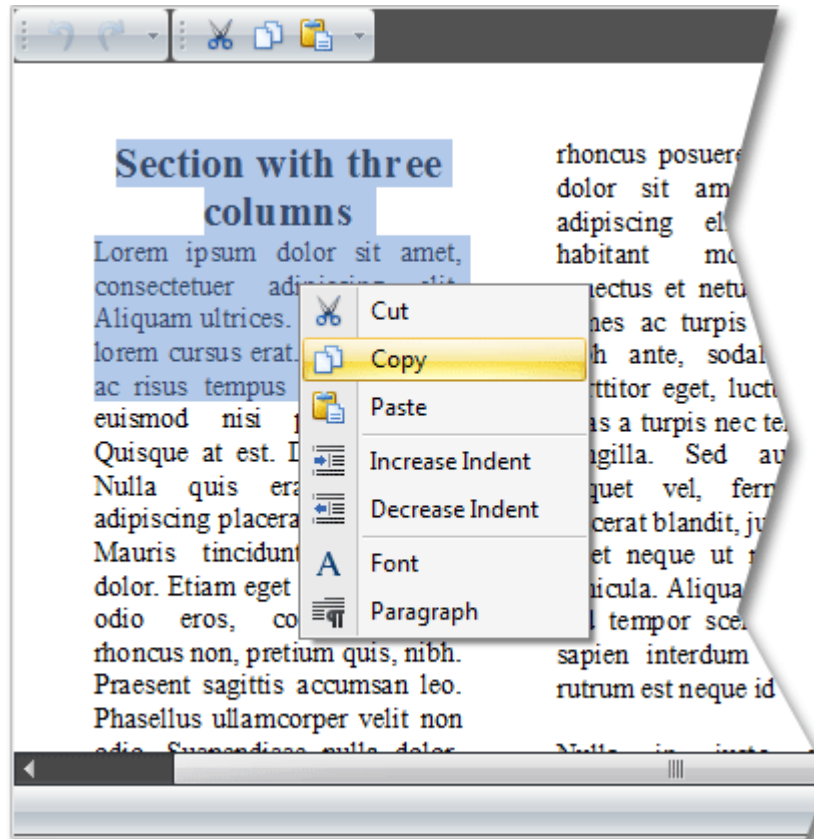


MDI tabs
Ribbon



Controls – DevExpress Winform

XtraRichEdit



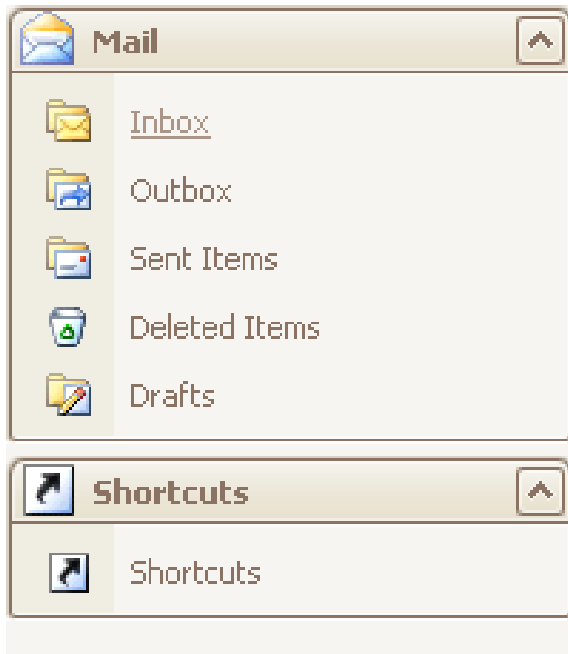
XtraCombo

Controls – DevExpress Winform

XtraNavBar

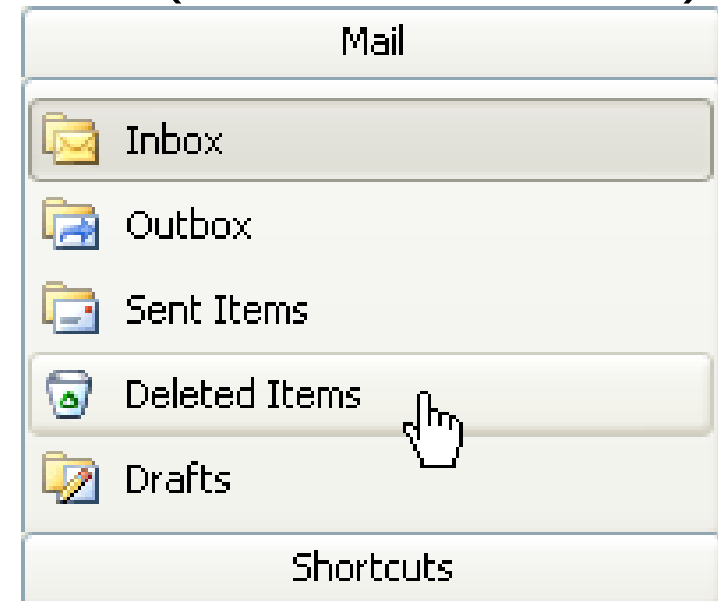


(Navigation bar)
(Ms. Outlook 2007)



(Explorer bar)
(Win Explorer)

(Side bar)
(Ms. Outlook 2003)



Controls – DevExpress Winform

XtraGrid

The image displays two DevExpress Winform controls. The top control is an XtraForm containing a vertical list of names (Nancy Davolio, Andrew Fuller, Janet Leverling) and a photo of Janet Leverling. The bottom control is an XtraGrid displaying a table of vehicle data, categorized by 'Truck' and 'Sports'.

Top Control (XtraForm):

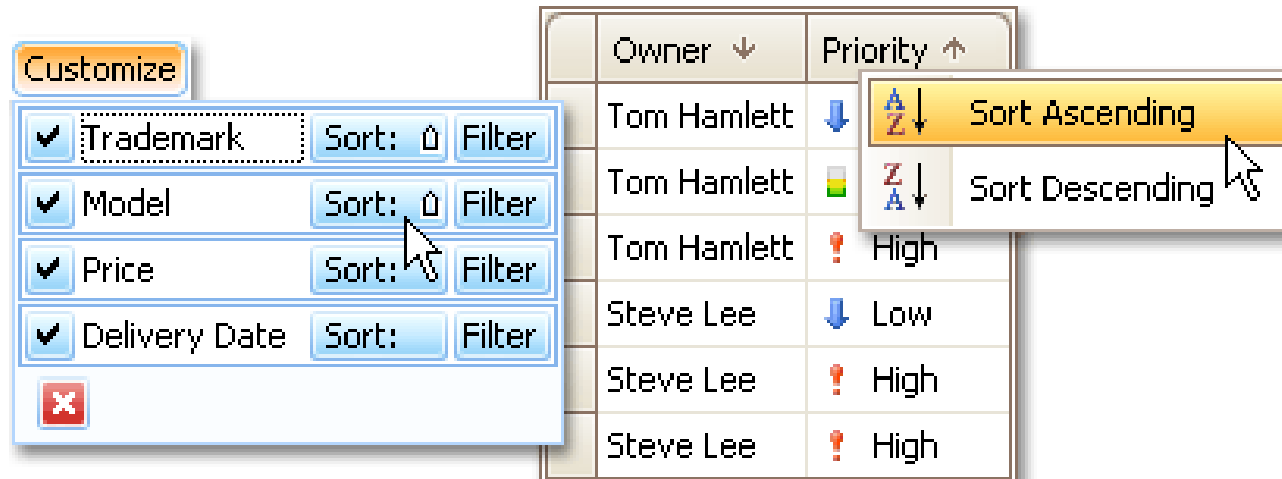
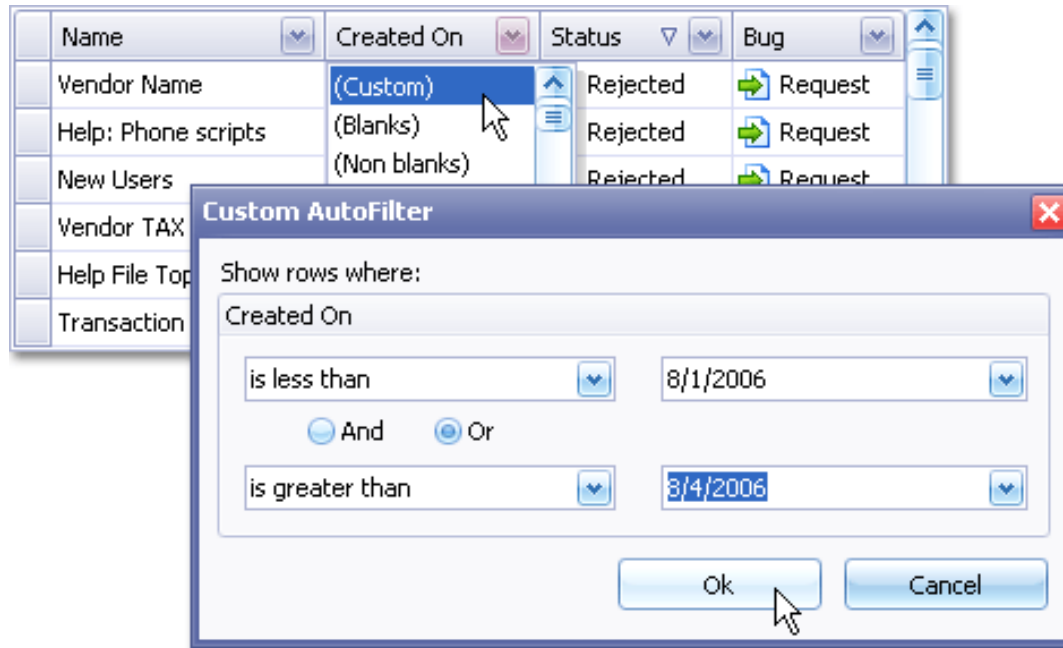
- Names: Nancy Davolio, Andrew Fuller, Janet Leverling
- Photo: Janet Leverling
- Courtesy title: Ms.
- Birth Date: 8/30/1963
- Home Phone: (206) 555-3412

Bottom Control (XtraGrid):

Main		Performance Attributes		Picture
Trademark	HP	Cyl		
Category	Model	Capacity	# of Gears	Icon
Price	Delivery D...	In Stock?	Automatic Transmission	
Category: Truck				
Ford		135	4	
Truck	Ranger FX-4	2.3 Liters	5	
\$12,565.00	3/15/2002	☒	☒ Yes ☐ No	
Nissan		143	4	
Truck	Crew Cab SE	2.4 Liters	4	
\$12,800.00	1/12/2002	☒	☒ Yes ☐ No	
Category: Sports				
☒ ([Cyl] = '4')				

Controls – DevExpress Winform

XtraGrid



Controls – Telerik RadControl

RadGridView

Furniture De Trevi

 **New**  **Delete**

Photo	ProductName	Manufacturer	Lining	Price	Condition	SalesRepresentative
	Wooden Table	Molzano Italy	wood	\$3,534.00	☒	Simonette Saylor
	White Leather Chair	Molzano Italy	polyester	\$436.00	☒	Bianka Mastoni
	Small Glass Table	L'Ouruje France	metal plates	\$3,480.00	☒	Marshal Duboa
	Red Leather Couch	Pront Italy	artificial le...	\$1,534.00	☒	Paolo Rossini
	Plywood Table	Spidea England	wood	\$234.00	☒	Silvester Williams
	Modern Design Ch...	Atech USA	leather	\$777.00	☒	Alan Smith

Product Details

Product Name:

Manufacturer:

Sales rep:

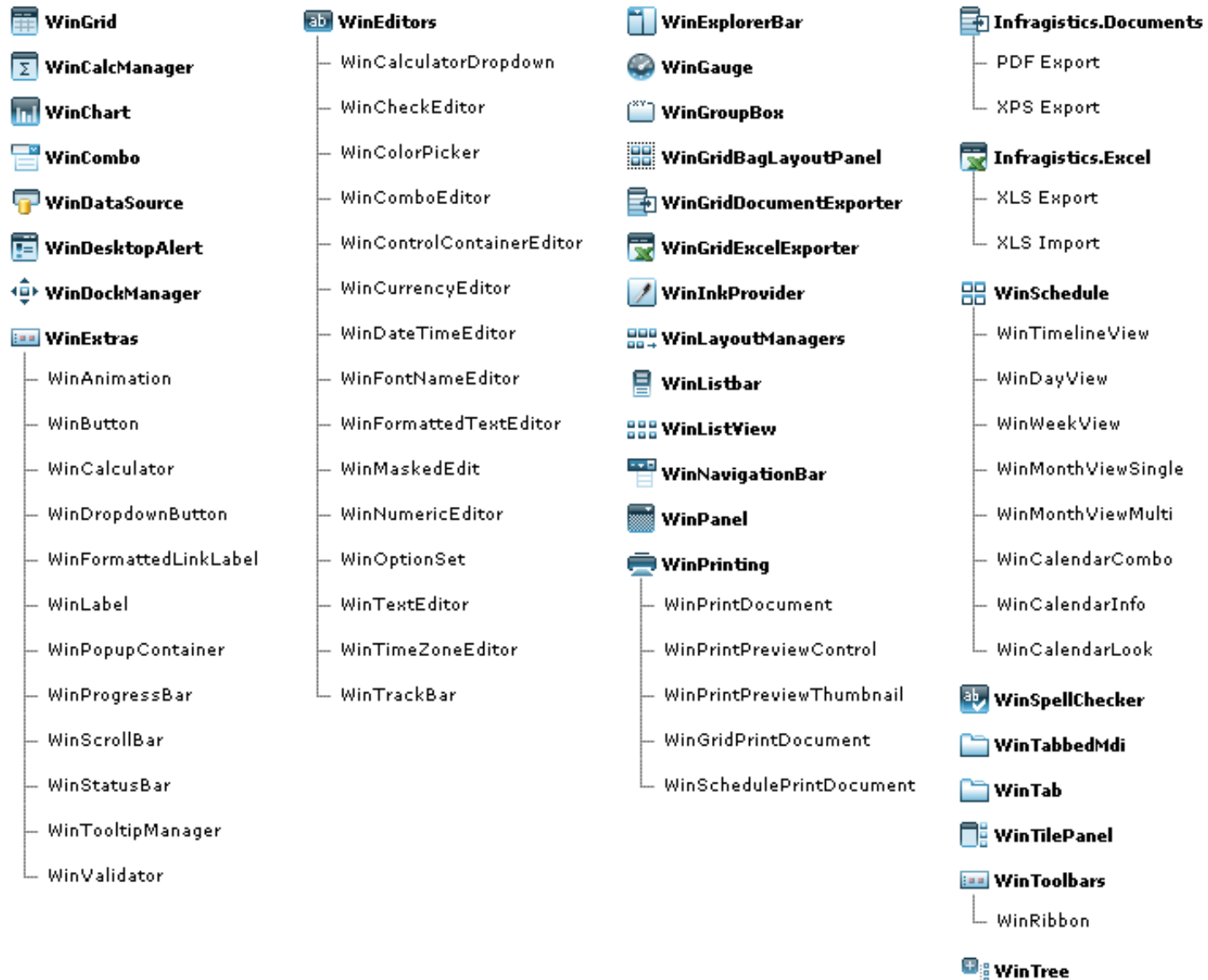
Lining:  Laura Callahan

Dimension:  Marshal Duboa

 Paolo Rossini

Controls – Infragistics Windows Forms Controls

Explore the many Windows Forms Controls and Components!



The more powerful & usable GUI components as well as technical solutions are presented, the higher quality & usability applications are required.

Reference

- ✦ <http://www.eclipse.org/swt/>
- ✦ http://en.csharp-online.net/CSharp_Certification%2C_Development%2C_and_Training
- ✦ <http://www.apl.jhu.edu/~hall/CWP-Chapter13/>
- ✦ <http://www.linuxjournal.com/article/2627> (extend Java widget)

C# user control

✦ Đọc & làm theo hướng dẫn:

- <http://www.java2s.com/Code/CSharp/GUI-Windows-Form/User-Control.htm>
- http://www.akadia.com/services/dotnet_user_controls.html

✦ User-control nhập địa chỉ Việt Nam:

- Nhập số nhà, tên đường
- Nhập / chọn phường, xã; quận huyện
- Nhập / chọn tỉnh, thành
- Link googlemap
- ...